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*J. Chem. Phys.* **32**, 1245 (1960); 10.1063/1.1730882

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# Unimolecular Decomposition of Chemically Activated *sec*-Butyl Radicals from H Atoms plus *cis*-Butene-2\*

B. S. RABINOVITCH AND R. W. DIESEN

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(Received September 23, 1958)

Chemically activated *sec*-butyl radicals have been produced at temperatures between  $-103^\circ$  and  $25^\circ\text{C}$  by reaction of H atoms with *cis*-butene-2. H atoms were allowed to effuse through a molecular slit into a spherical reactor containing olefine at low pressure. The system resembles the Polanyi diffusion flame, and some solutions for the radial distribution functions of the species involved are given. Disproportionation-recombination ratios for *sec*-butyl radicals, and for methyl plus butyl radicals, were measured as 0.94 and 0.30 respectively. The ratios of stabilization *vs* decomposition by C—C rupture for the energized butyl radicals were measured as a function of pressure and temperature. Hydrogen and helium were also used as inert gases.  $k_d$ , the observed rate constant for decomposition is  $\sim 10^7 \text{ sec}^{-1}$ . The energized butyl radicals are closely monoenergetic. The unimolecular decomposition behavior is compared with theory wherein expressions due to Marcus are employed. All vibrational degrees of freedom may be taken as active.

## INTRODUCTION

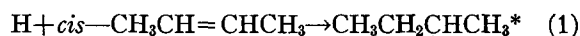
VIBRATIONALLY excited species can be formed by association and addition reactions of atoms and radicals. The association product may be a molecule<sup>1</sup> or a radical.<sup>2</sup> Unimolecular decomposition of the energized species produced by chemical reaction may involve rupture of the newly formed bond, or may proceed through rupture of some other bond. In either case, measurement of the rate of decomposition of the energetic species and of collisional deactivation is of interest and has an evident bearing on the study of unimolecular reactions.

Despite limitations in versatility as a means of producing vibrationally excited species, the present method is not without useful advantages. By employing different association or addition reactions the same association complex can be produced in different states of vibrational excitation at a particular temperature. Also, since the critical energy is supplied by bond formation it is possible to study reactions at low temperatures, with consequent reduced spread in the energy distribution of the decomposing species. Use of low temperatures has the practical advantage of reducing the importance of complicating secondary or side reactions. Finally, the chemically activated decomposition reaction may be studied over a wider range of temperatures than is generally feasible in conventional thermal studies.

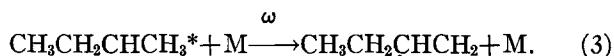
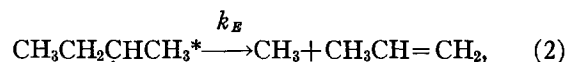
The present paper describes the reaction of energized *sec*-butyl radicals formed by addition of H atoms to *cis*-butene-2.

## Rationale of the Experiment

*Sec*-butyl radicals are produced homogeneously by reaction (1). The product radicals are energized with respect to their own ground



state by an amount,  $E$ , which has the minimum value,  $E_{\text{min}} = \Delta E_0' + E_H$ , where the quantities are defined in Fig. 1. Although unstable with respect to C—H rupture, the butyl radicals are also unstable with respect to C—C rupture by reaction (2), and possess excess energy,  $E^+$ , for process (2) of minimum amount,  $E_{\text{min}}^+ = E_{\text{min}} - E_0$ . The formed butyl radicals decompose essentially by C—C rupture.



The rate of (2) may be determined by the competition with the deactivation collision process (3). This takes into account the exponential form of the distribution function of collisions with time and assumes, conventionally, a similar "random" distribution in lifetimes of radicals of given energy.<sup>3,4</sup> It will be seen later that the formed radicals are closely monoenergetic. Determination of the average relative rates of (2) and (3) is made by comparison of the amounts of decomposition ( $D$ ) and stabilization ( $S$ ) products. The

\* Abstracted in part from the Ph.D. thesis of R. W. Diesen, University of Washington, 1958. This work was supported by the Office of Naval Research.

<sup>1</sup> R. A. Marcus, J. Chem. Phys. **20**, 352, 359 (1952), has collected and interpreted data dealing with simple paraffin hydrocarbon molecules.

<sup>2</sup> (a) Rabinovitch, Davis, and Winkler, Can. J. Research **B21**, 251 (1943); (b) A summary of some recent examples has been given by Pitts, Thompson, and Woolfolk, J. Am. Chem. Soc. **80**, 66 (1958).

<sup>3</sup> (a) N. B. Slater, Proc. Leeds Phil. Soc. **6**, 259 (1955), (b) J. Chem. Phys. **24**, 1256 (1956).

<sup>4</sup> It is assumed here that each collision leads to deactivation; i.e.,  $k_3 = \omega$ , where  $\omega$  is the mean frequency of collisions of butyl radicals at a particular total pressure; if the probability of deactivation were less than unity, say  $\eta$  (assumed constant), then  $k_3 = \eta\omega$ . We will return to this point later.

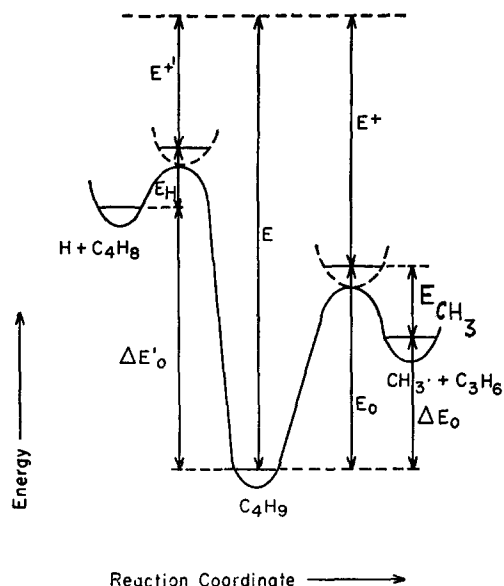
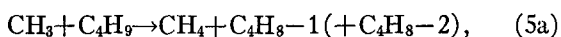
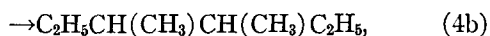
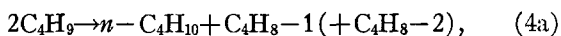


FIG. 1. Schematic potential energy diagram.

principal secondary reactions following (2) and (3) are



Further discussion of these and other reactions may be reserved until later.

Define  $k_a$  as the apparent or observed average rate constant for decomposition at a particular pressure. Then

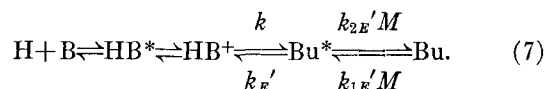
$$k_a = \omega \cdot D/S = \frac{\omega \int (k_E/k_E + \omega) f(E) dE}{\int (\omega/k_E + \omega) f(E) dE}, \quad (a)$$

where  $f(E)dE$  is the fraction of butyl radicals formed in the energy range  $E$  to  $E+dE$ , and the integral is over all relevant energies. Two important limiting forms for  $k_a$  are

$$\omega \rightarrow \infty, \quad k_{a\infty} = \frac{\int k_E f(E) dE}{\int f(E) dE} = \langle k_E \rangle \quad (b)$$

$$\omega \rightarrow 0, \quad k_{a0} = \frac{\int f(E) dE}{\int [f(E) dE/k_E]} = 1/\langle 1/k_E \rangle. \quad (c)$$

The general form of  $f(E)$  is now desired. When considering the thermal association reaction of hydrogen atoms with butene, it is easier to examine the microscopic reverse process, i.e., the decomposition of butyl radicals [reaction scheme (7)].



Consider a hypothetical equilibrium between H atoms (H) plus butene (B) and butyl radicals (Bu) in the high-pressure region for butyl radical decomposition. Then, as may be shown analytically,<sup>5</sup> and as is evident by application of the principle of detailed balancing, the rate of decomposition of energized butyl radicals (Bu)\* in any particular energy range is equal to the reverse process of formation. Thus the required form is

$$f(E) dE = \frac{k_E' K(E) dE}{\int_{E_{\min}}^{\infty} k_E' K(E) dE}, \quad (d)$$

where  $K(E) = k_{1E'}/k_{2E'}$ , the Maxwell-Boltzmann distribution function. The expression  $f(E)$  retains this form even at pressures where Bu decomposition to give H+B would no longer be in the high-pressure region. This follows since  $E_H$  is approximately 3 kcal, and the collision complex,  $\text{HB}^*$ , continues in reaction equilibrium with H+B at lower pressures.<sup>6</sup> The expression for  $f(E)$  also holds even though Bu actually decompose preferentially by another mechanism (C—C rupture).

The ratio,  $k_{a\infty}/k_{a0}$ , provides a useful measure of the energy spread of the formed radicals. If  $k_E$  is replaced by  $k_E'$  in Eqs. (b) and (c), i.e., if the decomposition process of interest were back reaction by C—H rupture as in (7), then the ratio takes on a simple form, namely  $k_{a\infty}/k_{a0} = \langle k_E'^2 \rangle / \langle k_E' \rangle^2 = F$ . The quantity  $F$ , defined by Johnston,<sup>7</sup> provides a measure of the spread of values of  $k_E'$  and hence also of the energy spread of the formed radicals. A similar interpretation, although not of such simple form, applies if  $k_E$  is retained in the ratio.<sup>†</sup>

## EXPERIMENTAL METHOD

### Materials

*cis*-Butene-2 (Phillips research grade) was further purified until it contained 0.020% *trans*-butene-2 and

<sup>5</sup> See R. W. Diesen, Ph.D. thesis, University of Washington, for details; this may be shown for any pressure.

<sup>6</sup> Cf. S. W. Benson and A. E. Axworthy, *J. Chem. Phys.* **21**, 428 (1953).

<sup>7</sup> H. S. Johnston, *J. Chem. Phys.* **20**, 1103 (1952); H. S. Johnston and J. R. White, *ibid.* **22**, 1969 (1954). The averages in  $F$  are now over the function  $K(E)$ .

<sup>†</sup> Then  $k_{a\infty}/k_{a0} = [(\langle k_E' \cdot k_E' \rangle \cdot \langle k_E'/k_E \rangle) / \langle k_E' \rangle^2]$  and this ratio may partake also of the meaning of Johnston's  $G$  quantity; e.g., if in some special case  $k_E' \sim \text{constant}$ ,  $k_{a\infty}/k_{a0} = \langle k_E \rangle \cdot \langle 1/k_E \rangle$ , i.e., of the nature of  $G$ .

0.003% propane. Tank hydrogen was passed through silica gel at  $-195^{\circ}$ . Helium was of 99.5% purity and after passing through silica gel at  $-195^{\circ}$  was found to be mass spectrometrically pure.

### Apparatus and Procedure

A Pyrex glass vacuum system was used. Atomic hydrogen, generated in a water-cooled Wood-Bonhoeffer type discharge tube at 0.1–0.6 mm pressure, effused through a rectangular slit, ( $0.3 \times 0.005$  mm with jaws ground to 0.01 mm edge), at the end of a finger located at the center of a spherical reaction flask through which *cis*-butene-2 at reaction pressure was passed. Two reactors, one 500 cc (B) and the other 1 liter (C), were used. The distance from slit to discharge region was approximately 1.5 cm. The flow of hydrogen, saturated with water at  $0^{\circ}\text{C}$  in a manostat, was measured with a bubble counter.

Butene-2 at a constant pressure in a reservoir burette was admitted to the reactor at a controlled rate (3–6 cc at NPT/hr). The reactor was pumped by a two-stage mercury diffusion pump. Flow out of the reactor was controlled by a diaphragm valve. Products were trapped at  $-195^{\circ}$  in front of the diffusion pump, in a silica gel trap at  $-195^{\circ}$  following the pump, while some hydrogen passed on into a bulb to be collected and measured. An appropriate gas handling line was used to separate products, whose pressures were measured in standard volumes by suitably isolated McLeod and Pearson gauges.

Runs were performed as follows. The flow of hydrogen and butene was adjusted and the discharge tube turned on. After a period of 1/2 to 1 hour (during which the species in the reactor came to a steady state) liquid nitrogen was placed around the collection traps, and the volume in the butene manostat and the start time noted. The duration of the runs ranged from 100 to 400 minutes.

The products were partially separated by distillation into four fractions.

### Analysis

All the products were identified by (a) comparison with known samples on characteristically different gas-liquid phase chromatographic columns, or (b) isolation by GLPC and identification by a Consolidated 21–103 mass spectrometer. All quantitative analysis was done by gas chromatography.

Fraction I (methane and hydrogen) was analyzed on a one meter silica gel column (60–80 mesh) at  $25^{\circ}\text{C}$ .

Fraction II ( $\text{C}_2$  and  $\text{C}_3$  hydrocarbons) was analyzed at  $25^{\circ}\text{C}$  on a column composed of 3 meters of glycol saturated with silver nitrate (50% by weight on 40–60 mesh firebrick) in series with 2 meters of *n*-butyl phthalate (40% by weight on Johns-Manville C-22 firebrick, 40–60 mesh).

Fraction III ( $\text{C}_3$ ,  $\text{C}_4$ ,  $\text{C}_5$  hydrocarbons) was analyzed at  $25^{\circ}\text{C}$  on a series column of 3 meters of glycol, saturated with silver nitrate, followed by 1 meter of *n*-butylphthalate. *Cis*-butene-2, came off last and did not tail over any of the compounds being analyzed. The butene-1 in this fraction was analyzed at  $25^{\circ}\text{C}$  on a column of 1 meter of dimethyl sulfolane (40% by weight) in series with 1 meter of *n*-butyl phthalate.

Fraction IV ( $\text{C}_6$  and above) was analyzed at  $100^{\circ}\text{C}$  on 2 meters of *n*-decyl phthalate (40% by weight). The absence of any high boilers in Fraction IV above  $\text{C}_8$  was confirmed by mass spectrographic analysis.<sup>8</sup>

### Homogeneity of Primary Reaction

It is important that the butyl radicals be formed in the gas phase. If formed at the wall the surface could act as an energy sink. Calculation of the radial H atom concentration distribution in butene is analogous to the problem of the Polanyi sodium diffusion flame, without streaming at the nozzle, for constant concentration of atmosphere gas. The steady-state equation of continuity has been discussed previously.<sup>9</sup>

When the fraction of H atoms that get to the wall is small, it is given by

$$\{(1 + Cr_w) \exp[-C(r_w - r_0)]\} / (1 + Cr_0),$$

where  $r_0$  is the effective radius of the effusion source,  $r_w$  is the radius of the reactor, and  $C^2 = kp_B/D_H$ . For  $r_w = 6.5$  cm, the fraction is <1% at butene pressures above 0.1 mm, 0.2 mm, and 0.3 mm at  $25^{\circ}$ ,  $-78^{\circ}$ , and  $-103^{\circ}$ , respectively. In two instances at lower pressure the fraction rose to 15%; even in these cases, however, reflection of H atoms at the wall<sup>10</sup> indicates the fulfillment of the requirement of homogeneous reaction.

### Back-Effusion Correction

Products collected were corrected for loss by back-effusion through the hydrogen entry slit. The ratios of the amounts of products to reactant butene collected are direct measures of their relative pressures in the reactor, if the concentration gradients in the reactor are the same. The radial concentration function for butene and propylene is shown in Appendix IA to be essentially a constant. This conclusion for propylene may be extended to all stable products since the generation of other species is less radially dependent (see Appendix IB) than propylene. The effusion corrections were made relative to the experimentally determined loss of butene reactant, weighted by the square root of the inverse molecular weight ratio.

<sup>8</sup> In preliminary runs, when care was not taken to eliminate all the mercury from the reactor, mercury dimethyl, mercury methyl butyl, and mercury dibutyl were found.

<sup>9</sup> J. F. Reed and B. S. Rabinovitch, J. Phys. Chem. **59**, 261 (1955).

<sup>10</sup> B. S. Rabinovitch and D. H. Dills, unpublished work. American Chemical Society Meeting, San Francisco (April, 1958).

TABLE I. Results with H atoms and *cis*-butene-2 at 25°C.

| Butene <i>p</i> , mm                                   | 0.050 | 0.10 | 0.20 | 0.39 | 0.83 | 1.18 |
|--------------------------------------------------------|-------|------|------|------|------|------|
| Butene reacted %                                       | 1.4   | 1.8  | 1.0  | 2.2  | 0.9  | 0.4  |
| Methane*                                               | 0.25  | 0.72 | 0.59 | 1.10 | 0.44 | 0.48 |
| Ethane                                                 | 2.5   | 8.0  | 3.25 | 5.10 | 1.0  | 0.20 |
| Ethylene                                               | 0.7   | ...  | 0.16 | 0.92 | ...  | 0.09 |
| Propane                                                | 0.01  | 0.05 | 0.11 | 0.11 | 0.09 | 0.03 |
| Propylene                                              | 7.0   | 18.8 | 9.0  | 14.4 | 2.65 | 1.31 |
| Isobutane                                              | 0.03  | 0.29 | 0.05 | 0.25 | ...  | ...  |
| <i>n</i> -Butane                                       | 0.09  | 0.32 | 0.33 | 1.15 | 0.63 | 0.38 |
| <i>t</i> -Butene-2                                     | 0.18  | 0.45 | 0.44 | 1.5  | 0.40 | 1.23 |
| Butene-1                                               | 0.11  | 0.47 | 0.49 | 1.1  | 0.52 | 0.25 |
| Isopentane                                             | 0.45  | 2.90 | 2.00 | 3.75 | 1.26 | 0.71 |
| 2,3-Dimethyl pentane                                   | ...   | ...  | 0.02 | 0.20 | ...  | ...  |
| 3,4-Dimethyl hexane                                    | ...   | 0.28 | 0.42 | 1.18 | 0.55 | 0.51 |
| <i>S/D</i>                                             | 0.12  | 0.25 | 0.44 | 0.62 | 1.47 | 2.08 |
| <i>k<sub>a</sub></i> 10 <sup>7</sup> sec <sup>-1</sup> | 0.83  | 0.80 | 0.91 | 1.22 | 1.13 | 1.14 |

\* The products are in units of 10<sup>-2</sup> cc of gas (NTP) and are the actual quantities found corrected for loss by effusion.

The corrections were in most instances less than 10% and ranged from zero at the lower pressures up to 50% for methane in one instance at highest pressure.

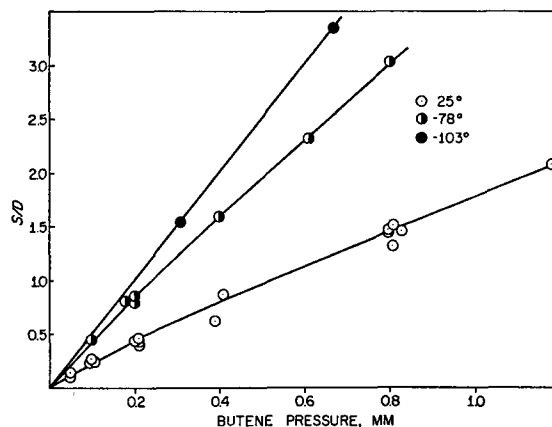
## RESULTS AND DISCUSSION

Tables I and II present results and product compositions obtained with pure *cis*-butene-2 in a typical run at each pressure, over a range of pressures at 25°, -78°, and -103°. From one to five runs were done at each pressure.

TABLE II. Results with H atoms and *cis*-butene-1 at -78° and -103°.

| Temp. °C                                               | -78    | -78  | -78  | -78  | -78  | -103 | -103 |
|--------------------------------------------------------|--------|------|------|------|------|------|------|
| Butene <i>p</i> , mm                                   | 0.10   | 0.20 | 0.40 | 0.61 | 0.80 | 0.31 | 0.67 |
| Butene reacted %                                       | 0.8    | 0.7  | 1.0  | 1.9  | 0.6  | 1.4  | 1.5  |
| Methane*                                               | (Lost) | 0.54 | 0.61 | 0.65 | 0.66 | 0.52 | 0.62 |
| Ethane                                                 | 0.94   | 2.1  | 1.39 | 1.54 | 0.27 | 1.9  | 0.87 |
| Ethylene                                               | 0.16   | 0.24 | 0.11 | 0.03 | 0.03 | 0.12 | 0.01 |
| Propane                                                | 0.15   | 0.06 | 0.07 | 0.07 | 0.04 | 0.06 | 0.02 |
| Propylene                                              | 3.1    | 5.9  | 5.18 | 3.91 | 1.19 | 4.36 | 2.66 |
| Isobutane                                              | ...    | 0.02 | ...  | 0.07 | ...  | 0.05 | ...  |
| <i>n</i> -Butane                                       | 0.27   | 0.68 | 1.35 | 1.49 | 0.59 | 1.50 | 1.56 |
| <i>t</i> -Butene-2                                     | 0.73   | 0.68 | 0.71 | 0.71 | 0.47 | 0.70 | 0.57 |
| Butene-1                                               | 0.28   | 0.66 | 0.81 | 0.97 | 0.34 | 1.12 | 1.07 |
| Isopentane                                             | 0.70   | 1.78 | 2.09 | 2.61 | 0.68 | 1.31 | 1.43 |
| 2,3-Dimethyl pentane                                   | ...    | ...  | ...  | ...  | ...  | 0.12 | ...  |
| 3,4-Dimethyl hexane                                    | 0.13   | 0.79 | 1.60 | 1.61 | 0.64 | 1.34 | 1.95 |
| <i>S/D</i>                                             | >0.44  | 0.85 | 1.60 | 2.32 | 3.03 | 1.55 | 3.35 |
| <i>k<sub>a</sub></i> 10 <sup>7</sup> sec <sup>-1</sup> | <0.56  | 0.58 | 0.62 | 0.65 | 0.65 | 0.53 | 0.53 |

\* See footnote, Table I.

FIG. 2. The ratio *S/D* as function of *cis*-butene-2 pressure.

The products are similar to those found by Frey and Hepp<sup>11</sup> in some remarkably thorough analytical work on the pyrolysis of di-*sec*-butyl mercury at 500°. In two or three runs with regard to *trans*-butene-2 product, and in one run for ethylene product, the amounts of these products were somewhat anomalous.

Figure 2 summarizes the *S/D* ratios obtained in all runs with pure butene. In Figs. 3 and 4, respectively, plots of *k<sub>a</sub>* vs *p* and *k<sub>a</sub>* vs 1/*p* are extrapolated to obtain values of *k<sub>a0</sub>* and *k<sub>a∞</sub>*.

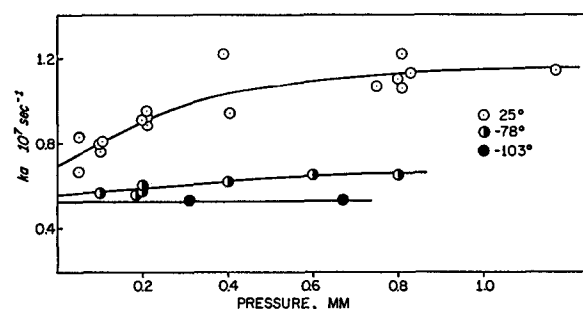
Table III summarizes the *S/D* ratios obtained in the presence of inert gases at different pressures, at two temperatures. The equivalent pure reactant pressures at the same *S/D* given in column 6 of this table were found from Fig. 2, by extrapolation in a few cases.

Reference 5 may be consulted for details of all runs, product composition, etc.

## Calculation of *S/D*. Reaction Mechanism

In the calculation of *S/D* the following definitions were employed:

$$\begin{aligned}
 S &= \text{isopentane} + \text{methane} + 2(n\text{-butane}) \\
 &\quad + 2(3,4\text{-dimethylhexane}) + 2,3\text{-dimethylpentane} \\
 D &= \text{propylene} + \text{propane} + \text{ethylene} + \text{isobutane} \\
 &\quad + 2,3\text{-dimethylpentane.}
 \end{aligned}$$

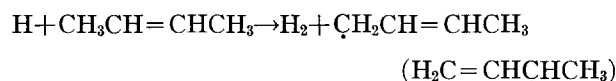
FIG. 3. Plot of *k<sub>a</sub>* vs pressure extrapolated to zero pressure.

<sup>11</sup> F. E. Frey and H. J. Hepp, J. Am. Chem. Soc. **55**, 3357 (1933).

These equations are based upon the consideration that reactions 1–6 do indeed predict the major products found (Tables I and II). It remains to show the origin of all the products and the reasons why complicating reactions are not of importance.

### The Primary Process

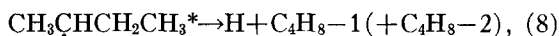
In addition to reaction (1), H abstraction could occur. The collision efficiency of reaction (1) has been



determined<sup>12</sup> at room temperature as  $9 \times 10^{-4}$ . That for abstraction reaction between D atoms and butene-2 (of undetermined isomeric composition) has been given<sup>13</sup> as  $6.7 \times 10^{-5}$  with a possible correction downward<sup>14</sup> to  $3 \times 10^{-6}$ . Abstraction by H is even less favored.<sup>14</sup> At lower temperatures abstraction is again still less favored. In fact, only a trace of pentene and no octene products, as would arise by combination of methallyl with methyl or butyl, were found experimentally at any temperature; this supports Trotman-Dickenson's correction and places the upper limit of the relative extent of abstraction to addition at  $<1\%$  at room temperature or below.

### Secondary Processes

Possible reactions in addition to (2) and (3) include back reaction and rearrangement. Reaction (8) involves



a critical energy larger than that for (2); in view of the greater molecular complexity, the rate of (8) would be expected to be appreciably less than the rate of the analogous reaction of  $\text{C}_2\text{H}_5^*$ , which is a little less than  $k_a$  observed here.<sup>10</sup> (8) cannot be ruled out

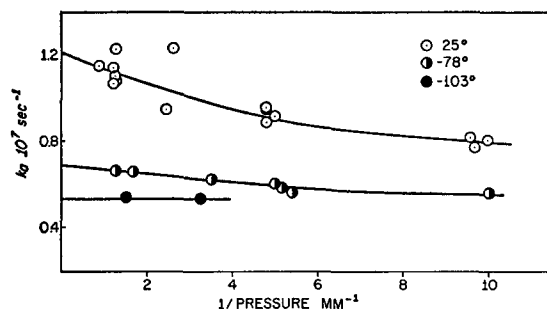


FIG. 4. Plot of  $k_a$  vs  $1/\text{pressure}$  extrapolated to infinite pressure.

<sup>12</sup> H. W. Melville and J. C. Robb, Proc. Roy. Soc. (London) **A196**, 445 (1949); **A202**, 181 (1950).

<sup>13</sup> B. de B. Darwent and R. Roberts, Discussions Faraday Soc. **14**, 55 (1953).

<sup>14</sup> A. F. Trotman-Dickenson, *Gas Kinetics* (Academic Press, Inc., New York, 1955).

TABLE III. Relative inert gas-butene collision efficiencies,  $\eta$ , on a collision-per-collision basis.

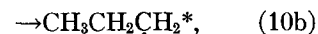
| Inert gas      | Temp. | Reactant press., mm | Inert press., mm | $S/D$ | Equival. pure reactant same $S/D$ , mm | $\eta$   |
|----------------|-------|---------------------|------------------|-------|----------------------------------------|----------|
| H <sub>2</sub> | 25    | 0.10                | 0.40             | 0.79  | 0.40                                   | 0.39     |
|                |       | 0.14                | 0.66             | 0.91  | 0.46                                   | 0.25     |
|                |       | 0.12                | 0.69             | 1.04  | 0.55                                   | 0.32     |
|                |       | 0.11                | 1.09             | 1.04  | 0.55                                   | 0.21     |
|                |       | 0.14                | 1.86             | 1.77  | 1.0                                    | 0.27     |
|                |       | 0.17                | 3.8              | 3.0   | 1.8                                    | 0.22     |
|                |       |                     |                  |       |                                        | Av. 0.28 |
| He             | 25    | 0.12                | 0.68             | 0.78  | 0.38                                   | 0.29     |
|                |       | 0.13                | 1.48             | 0.93  | 0.48                                   | 0.18     |
|                |       |                     |                  |       |                                        | Av. 0.24 |
| H <sub>2</sub> | -78   | 0.26                | 0.54             | 2.90  | 0.77                                   | 0.48     |
|                |       | 0.26                | 1.34             | 4.64  | 1.3                                    | 0.39     |
|                |       | 0.30                | 2.70             | 6.74  | 1.9                                    | 0.33     |
|                |       |                     |                  |       |                                        | Av. 0.40 |

entirely on the basis of any crucial diagnostic test provided by the observed products<sup>15</sup>; it helps explain the occasional anomaly, previously mentioned, in the amount of *trans*-butene-2 found, as well as the fact that *trans*-butene-2 product exceeded the amount predicted from disproportionation reaction discussed below. However, (8) is negligible relative to (2) since the amounts of butene-1 and *trans*-butene-2 were only of the magnitude expected from the sequence (2)–(6), and were much less than the propylene product.

Reaction (9) would lead to the isolation of *n*-pentane, methyl heptane and *n*-octane which were not found. Decomposition of a small amount of *n*-butyl could however account for some ethylene product<sup>15</sup> (next section).

### Propylene Reaction

Propylene may react further with H to give propyl radicals, with reaction



Eq. (10a) predominating.<sup>16</sup> The competing reaction of H abstraction to produce allyl is only one-fifth as fast<sup>13</sup> and may be considered as negligible here; in fact, combination of butyl with allyl would yield heptene-1

<sup>15</sup> Compare an analogous reaction of isopropyl radical observed by C. A. Heller and A. S. Gordon, J. Phys. Chem. **62**, 1709 (1958).

<sup>16</sup> W. J. Moore, J. Chem. Phys. **16**, 916 (1948); Bradley, Melville, and Robb, Proc. Roy. Soc. (London) **A236**, 339 (1956) find 7.5% addition by (10b).

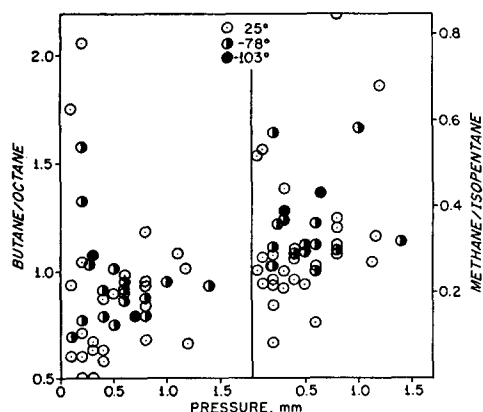
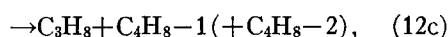
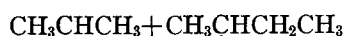
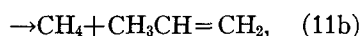
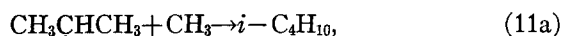


FIG. 5. Disproportionation-recombination ratios for butyl plus butyl and methyl plus butyl.

which was not detected. The chief reactions of isopropyl explain



the isobutane and 2,3-dimethyl pentane found. For the reason presented below, the extent of (11b) is estimated as two-tenths (11a). The extent of (12c) is equated to the amount of propane found. Employing the reasoning of Kraus and Calvert,<sup>17</sup> the extent of (12b) is six-fifths (12c).

*n*-Propyl radicals formed in (10b) may decompose to yield ethylene product<sup>3a</sup> although the extent of this process does not explain the total ethylene. The (small) amount found evidently also arose in another way.

It should be kept in mind that secondary reaction of propylene was only 1–2% of propylene formed, except for two runs.

#### Disproportionation and Recombination Reactions of Methyl and *sec*-Butyl Radicals

Reactions (4a) and (5a) yield *n*-butane and methane, with concomitant amounts of butenes. These saturated products rather than the olefins were used as a measure of stabilization since the *cis*-butene-2 formed could not be determined, while the *trans*-butene-2 contained the "anomaly" mentioned. In addition, the extent of reactions (12b) and (12c) is adequately included at the same time by use of the *n*-butane figure. A negative correction for (11b) should be made to the methane in estimating stabilization.

<sup>17</sup> J. W. Kraus and J. E. Calvert, J. Am. Chem. Soc. **79**, 5921 (1957).

All data for the determination of the disproportionation-recombination ratios  $k_{4a}/k_{4b}$  and  $k_{5a}/k_{5b}$  are summarized in Fig. 5. The numbers do not include products formed in reactions (11b) and (12b). The average for the first ratio is 0.95, independent of temperature and pressure within experimental error. Existing values are  $\sim 0.7$ ,<sup>18</sup> 0.61,<sup>19(a)</sup> and 2.3.<sup>17</sup> The scatter in Fig. 5 is attributed mainly to analytical error. The average ratio  $k_{5a}/k_{5b}$  for the cross-reaction of methyl and *sec*-butyl is 0.30, also independent of temperature and pressure. A value from Frey and Hepp<sup>18</sup> is  $\sim 0.3$ . Since the corresponding ratio for  $\text{CH}_3 + \text{C}_2\text{H}_5$  is<sup>19(b)</sup>  $\sim 0.06$ , which leads to 0.12 for  $k_{11b}/k_{11a}$ , the ratio  $k_{11b}/k_{11a}$  was set noncommittally at 0.2, intermediate between 0.12 and 0.31.

The relative amounts of butene-1 and butene-2 formed in (4a) and (5a) may be found from the relationship between these

$$(\text{C}_4\text{H}_8 - 1) = \beta(n\text{-C}_4\text{H}_{10}) + \gamma(\text{CH}_4)$$

products (i.e., total of these products, corrected for reactions (11) and (12), but not for (8) if any). A plot of  $(\text{C}_4\text{H}_8 - 1)/(\text{CH}_4)$  vs  $(n\text{-C}_4\text{H}_{10})/(\text{CH}_4)$  shown in Fig. 6 yields  $\beta$  and  $\gamma$  from the slope and intercept;  $\beta = 0.5$ ,  $\gamma = 0.45$ . Then  $\text{C}_4\text{H}_8 - 2/\text{C}_4\text{H}_8 - 1 = 1.2$  in (5a) and 1.0 in (4a). Kraus and Calvert<sup>17</sup> have determined the latter ratio as 0.66 and correlate their number with the ratio of the number of abstractable hydrogens (2:3) which lead to the two products. Since  $D(\text{C}-\text{H})$  for primary hydrogen is greater than for secondary hydrogen, 0.67 would be a lower limit (apart from entropy factors), although quite possibly realized.

The fact that the total amount of *trans*-butene-2 found frequently exceeded that predicted from disproportionation (keeping in mind the lesser amount<sup>17</sup> of *cis*-butene-2 formed also and not measured) reflects its formation by other processes such as (8). On this

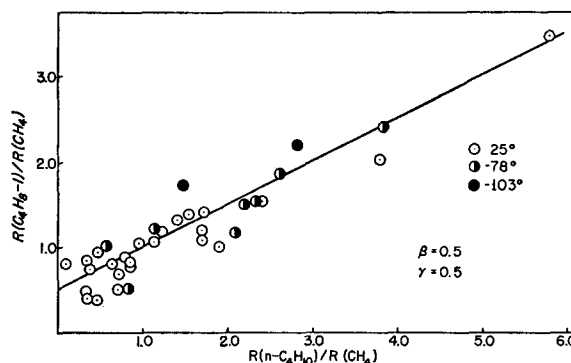


FIG. 6. Rate ratio relationship for butene-1.

<sup>18</sup> Calculated from the data of reference 11.

<sup>19</sup> (a) J. T. Gruver and J. G. Calvert, J. Am. Chem. Soc. **78**, 5208 (1956); (b) M. H. J. Wijnen, J. Chem. Phys. **22**, 1631 (1954); P. Ausloos and E. W. Steacie, Can. J. Chem. **33**, 1062 (1955).

basis little butene-1 was formed by (8) since it did not arise in such excess.

In Appendix IB the radial distribution function for butyl and methyl radical concentrations is discussed. It is shown there that the functional dependence for these radicals is less than  $1/r^2$ , and quite possibly concentrations may be assumed constant. The corresponding steady-state pressures vary between  $5 \times 10^{-6}$ – $5 \times 10^{-7}$  mm, depending on conditions. Two remarks may now be made with respect to the disproportionation-recombination ratios cited above. The first is that although butyl radicals are highly energized on formation, they make an average of  $10^5$ – $10^6$  collisions (depending on total butene pressure) with other molecules to every collision with another radical. Hence they should be appreciably thermalized before they react.<sup>20</sup> The second matter involves the fact that the radical concentration is finite at the wall. The number of reflections from the wall necessary to ensure that these reactions take place homogeneously may be estimated with the use of the equation in the displacement,  $l$ , after reflection,  $\langle l^2 \rangle = 6Dt$ , and is roughly 12 and 25 for butyl and methyl, respectively, depending on reaction conditions.<sup>5</sup> It was found at 0.01 mm ethylene at 25° that H atoms make many collisions before reacting at the wall,<sup>10</sup> so that the above disproportionation-recombination data could probably represent homogeneous reaction entirely.<sup>21</sup> Whether it does or not of course in no way affects  $S/D$ .

#### Other Reactions. $H + \text{Radical}$

Association of H atoms with butyl and methyl could give rise to butane and methane.  $k_{H+Bu}$  was estimated by assuming  $P=0.1$ ,  $E=0$ . The rate of formation of butane is

$$R_{H+Bu} = 4\pi \int_{r_0}^{r_w} k_{H+Bu} \cdot p_H \cdot p_{Bu} r^2 dr,$$

where  $p_H = b_0 e^{-C/r} / 4\pi D_H r$ . The calculated percentage of the observed butane product formed in this way varied from zero up to 5% for the worst case. *Ad hoc* support for the conclusion that this reaction is indeed negligible comes from the observed actual amounts of butane relative to butenes and octane. In any case, the effect of this reaction on  $S/D$  is insignificant. The amount of methane formed in the analogous fashion would be even less, since this association reaction is presumably in the falloff region at these pressures (cf. Appendix IB), while methyl partial pressure was less than that of butyl except at the lower pressures.

<sup>20</sup> (a) M. Boudart and J. T. Dubois, J. Chem. Phys. **23**, 223 (1955); (b) Bradley, Melville, and Robb, Proc. Roy. Soc. (London) **A236**, 318 (1956).

<sup>21</sup> The reaction of H atoms at the wall presumably represents addition to ethylene primarily; as discussed in the next section, addition of radicals to the double bond is known to be appreciably less efficient than is H addition.

#### Radical + Butene-2

For methyl radicals plus butene-2, both hydrogen abstraction and addition are possible. Since abstraction by H atoms is negligible, so might be also abstraction by  $\text{CH}_3$ , for which the specific rate constant (calculated as  $k_{\text{abst}} = 4.5 \times 10^5$  cc mole<sup>-1</sup> sec<sup>-1</sup> at 25°, from the data of reference 14) is less; this is confirmed by the absence of pentene and octene products. Now the rate of methyl addition to *cis*-butene-2 relative to hydrogen abstraction has been found<sup>22</sup> to be unity at 55°. The rate of addition of methyl to butene-2 is thus similarly negligible, as confirmed again by the absence of higher products (only trace amounts hexane and no nonane). These reactions cannot compete with reactions (5) and (6); for example,  $k_6 = 2 \times 10^{13}$  cc mole<sup>-1</sup> sec<sup>-1</sup>,<sup>23</sup> so  $k_6/k_{\text{abst}} = 4.5 \times 10^7$  while, as shown above,  $p_B/p_{\text{CH}_3} \sim 10^6$  only. The calculated maximum extent of addition or abstraction by methyl radicals under the worst conditions is 3% of the methyl radicals.

For similar reasons, hydrogen abstraction from butene-2, or addition to the double bond, by butyl radicals is also negligible.

Attack by these radicals on molecular hydrogen, arising in the reactor from the discharge or added as inert gas, is even slower than their reaction with butene-2.<sup>24</sup>

We conclude that any discrepancies with the above mechanism and deductions are minor in nature for present purposes and do not significantly affect the calculated values of  $S/D$  and  $k_a$ .

#### Thermochemical Desiderata

To calculate  $k_a$  some zero point energy differences in Fig. 1 must be estimated.<sup>25</sup> Now  $E_{\text{min}}$  is the sum of  $E_H$  and  $\Delta E_0'$ .  $E_H$ , the activation energy at 0°K, must be estimated from the collision yield quoted earlier,  $9 \times 10^{-4}$  at room temperature. The value of  $P$  for reaction (1) is less than unity,<sup>13</sup> and we have adopted  $E_H = 3$  kcal. Evaluation of  $\Delta E_0'$  depends upon knowledge of the energy of formation of *sec*-butyl radical, which in turn depends upon  $D(\text{sec}-\text{C}_4\text{H}_9-\text{H})$ . For some time the value of  $D(n-\text{R}-\text{H})$ , as in propane and butane, was believed to be of the order of, or less than,  $D(\text{CH}_3\text{CH}_2-\text{H})$ ; the value of  $D(\text{sec}-\text{R}-\text{H})$  is taken as 4–6 kcal/mole less.<sup>26</sup> More recently, Stevenson<sup>27</sup> determined  $D(n-\text{C}_3\text{H}_7-\text{H}) = 99$  kcal,  $D(\text{sec}-\text{C}_3\text{H}_7-\text{H}) = 94$  kcal and  $D(n-\text{C}_4\text{H}_9-\text{H}) = 101$  kcal/

<sup>22</sup> R. P. Buckley and M. Szwarc, Proc. Roy. Soc. (London) **A240**, 396 (1957).

<sup>23</sup> A. Shepp, J. Chem. Phys. **24**, 939 (1956).

<sup>24</sup> E. W. R. Steacie, *Atomic and Free Radical Reactions* (Reinhold Publishing Company, New York, 1954).

<sup>25</sup> Unfortunately the estimation of several quantities, especially  $\Delta E_0'$ , and  $\Delta E_0$ , is quite uncertain and in arriving at the order of magnitude to be used it is scarcely necessary to distinguish between heat and energy quantities, or to correct between room temperature and absolute zero for those quantities for which the difference largely cancels. The numbers calculated refer to 0°K.

<sup>26</sup> J. S. Roberts and H. A. Skinner, Trans. Faraday Soc. **50**, 339 (1949).

<sup>27</sup> D. P. Stevenson, Trans. Faraday Soc. **49**, 867 (1953).



mole. These values supported the value  $D(n\text{-C}_3\text{H}_7\text{-H})=100$  kcal from the pyrolysis of *n*-butyl benzene.<sup>28</sup> It now appears that this latter value is too high by 5–7 kcal, since it was based on an erroneous value of  $D(\text{C}_6\text{H}_5\text{CH}_2\text{-H})$ .<sup>29</sup> In the face of this uncertainty we have elected  $D(\text{sec-C}_4\text{H}_9\text{-H})=92$  kcal. Coupled with other known quantities,<sup>30</sup>  $\Delta E_0'=-38$  kcal/mole is obtained. Then  $E_{\text{min}}=41$  kcal.

The uncertainty in  $D(\text{sec-C}_4\text{H}_9\text{-H})$ , while affecting the magnitudes of  $\Delta E_0'$ ,  $E$ , and  $\Delta E_0$ , does not enter into the determination of the much more sensitive magnitude,  $E^+$ . For reaction (2),  $E_{\text{min}}^+=E_{\text{min}}-E_0$ , where  $E_0=\Delta E_0+E_{\text{CH}_3}$ . Using  $D(\text{CH}_3\text{-H})=102$  kcal and other known quantities,<sup>30</sup>  $\Delta E_0=25$  kcal. Mandelcorn and Steacie<sup>31,22</sup> found 6 kcal for addition of methyl to propylene. Now, the ratio of terminal/nonterminal addition may be inferred as 6/1 for this reaction at 235°,<sup>32</sup> while for methyl addition to butene-1 at 400° the corresponding ratio is 7/1.<sup>33</sup> Hence the 6 kcal value does correspond to  $E_{\text{CH}_3}$  for reaction (2) and  $E_0=31$  kcal<sup>34</sup>; then  $E_{\text{min}}^+=10$  kcal.

### Calculation of $k_a$

For the evaluation of  $k_a$ , or its limiting forms given in Eqs. (b) and (c), it is necessary to obtain expressions for  $k_E$ ,  $k_E'$ , and  $K(E)$ . Marcus<sup>1</sup> has developed equations using absolute rate theory<sup>35</sup> and we will employ his treatment, using much the same nomenclature.

A general expression from Marcus for  $k_E$  in terms of the vibrational and rotational degrees of freedom of the molecule is

$$k_E = P_1^+ N_E^+ / h P_1^* N_E^*, \quad (e)$$

where  $P_1^+$  and  $P_1^*$  are products of partition functions for adiabatic degrees of freedom of the activated complex and energized molecule respectively (e.g., over-all rotations).<sup>1</sup> We assume for the present no inactive internal degrees of freedom.  $N_E^+$ , the total number of internal rotational and vibrational energy levels at the total energy,  $E^+$ , for the activated complex is

$$N_E^+ = [P_R^+ / (r/2)!] \sum_{E_v^+=0}^{E^+} [(E^+ - E_v^+) / RT]^{r/2} P(E_v^+), \quad (f)$$

<sup>28</sup> C. H. Leigh and M. Szwarc, *J. Chem. Phys.* **20**, 407 (1952).  
<sup>29</sup> S. W. Benson and J. H. Buss, *J. Phys. Chem.* **61**, 104 (1957);  
 A. H. Schon and M. Szwarc, *Ann. Rev. Phys. Chem.* **8**, 447 (1957).  
<sup>30</sup> F. D. Rossini, *Selected Values of Physical and Thermodynamic Properties of Hydrocarbons* (Carnegie Press, Pittsburgh, 1953).

<sup>31</sup> L. Mandelcorn and E. W. R. Steacie, *Can. J. Chem.* **32**, 79, 474 (1954).

<sup>32</sup> Rust, Seubold, and Vaughan, *J. Am. Chem. Soc.* **70**, 95 (1948).

<sup>33</sup> J. R. McNesby and A. S. Gordon, *J. Am. Chem. Soc.* **79**, 5902 (1957).

<sup>34</sup> This value is higher than that measured above 275° by S. Bywater and E. W. R. Steacie, *J. Chem. Phys.* **19**, 172 (1951), namely 23 kcal; it does not seem possible to reconcile the latter quantity by any reasonable variation on  $D(\text{sec-C}_4\text{H}_9\text{-H})$ .

<sup>35</sup> Cf. Rosenstock, Wallenstein, Wahrhaftig, and Eyring, *Proc. Natl. Acad. Sci. U. S.* **38**, 667 (1952).

while the energy density expression for internal rotational and vibrational levels of energized molecules at the energy  $E$  is

$$N_E^* = [P_R^* / (r/2-1)! RT] \sum_{E_v=0}^E [(E - E_v) / RT]^{r/2-1} P(E_v), \quad (g)$$

where  $P_R$  is a partition function product for  $r$  active internal rotational degrees of freedom. When the energy is large,  $P(E_v)$ , the number of vibrational states at the energy  $E_v$ , can be expressed to good approximation in terms of a semiclassical expression for density of vibrational energy levels,<sup>36</sup> and with integration over the energy range we have

$$\sum_{E_v=0}^E P(E_v) = \int_0^E [E_v + E_z]^{n-1} dE_v / (n-1)! \prod_{i=1}^n h\nu_i, \quad (h)$$

where  $E_z$  is the zero point energy of the  $n$  vibrational degrees of freedom of frequency  $\nu_i$  ( $E_v > h\nu_i$  for all  $i$ ).

Similar primed expressions apply for  $k_E'$ .

The value of  $K(E)$  is given by

$$K(E) = \frac{N_E^{*'} e^{-E/RT}}{\int_0^\infty N_E^{*'} e^{-E/RT} dE}.$$

Substitution of these expressions into Eq. (d) gives

$$f(E) dE = \frac{N_E^{+'} e^{-E/RT} dE}{\int_{E_{\text{min}}}^\infty N_E^{+'} e^{-E/RT} dE}, \quad (i)$$

where all constants in  $k_E'$  and  $K(E)$  cancel.

To evaluate Eqs. (e) and (i) it is necessary to specify the model of the activated complex for the reverse association reaction (1) and the decomposition reaction (2), as well as for the butyl radical. The possible<sup>13,14</sup> values of the entropy of activation for reaction (1) suggest the choice of a "semirigid" complex. A double bond torsion of reduced frequency is assumed, and two new C—H bending frequencies at roughly half their normal frequencies are introduced. The remaining vibration frequencies are taken as in butene.<sup>37</sup> For ease of calculation the values actually used were rounded (in  $\text{cm}^{-1}$ ): 300 (2), 600, 675 (3), 1000 (7), 1300 (4), 1450 (5), 2950 (8). Two internal rotational degrees of freedom of the terminal methyl groups are taken as active and separable from over-all rotations. Anticipating later discussion, the calculated value of  $k_a$  is rather insensitive to the choice of model for the association complex, i.e., whether loose or rigid, details of vibration frequency assumptions, etc. and

<sup>36</sup> R. A. Marcus and O. K. Rice, *J. Phys. & Colloid Chem.* **55**, 894 (1951).

<sup>37</sup> J. E. Kilpatrick and K. S. Pitzer, *J. Research Natl. Bur. Standards* **38**, 191 (1947).

the above may be considered as a useful working model.

Since the entropy of activation for reaction (2) is low,<sup>14,31</sup> ( $P_{-2}$  factor  $\sim 10^{-4}$ ) a "rigid" complex is assumed with no internal degrees of freedom of "tumbling" of  $\cdot\text{CH}_3$  and  $\text{CH}_3\text{CH}=\text{CH}_2$  about axes transverse to the bond direction. For the present, all thirty vibration frequencies of the complex are taken the same as those in the butyl radical (one C—C stretching mode lost). The vibration frequencies used for butyl are listed in Appendix II. To simplify the computation only two internal degrees of rotational freedom were treated as such, namely those for terminal methyl rotation.<sup>38</sup> The hindering barriers around the central C—C bond in butane vary from  $\sim 3.5$  kcal up to 6–8 kcal for the major barrier and the values are taken to be the same in butyl $\ddagger$ ; incipient double bond formation in the decomposition complex will tend to raise these barrier heights, so this degree of freedom is treated as a low-frequency torsional vibration for both complex and butyl radical.<sup>§</sup>

Now in Eq. (f') for the reverse of the association reaction,  $P(E_v^{+'})$  is evaluated directly by inspection, as is both necessary and feasible since  $E^{+}$  is small. For the decomposition process, however,  $E^{+} \geq 10$  kcal and  $P(E_v^{+})$  is a large number, so that direct counting is not practical and Eq. (h) is not valid at these low energies just above C—H frequencies, although adequate for  $P(E_v)$ . A further approximation of the form

$$\sum P(E_v^{+}) = \int [E_v^{+} + aE_z^{+}]^{n-1} dE_v^{+} / (n-1)! \prod_{i=1}^n h\nu_i,$$

where  $a$  is an empirical constant,  $\sim 0.8$ , yields the right order of magnitude for  $P(E_v^{+})$  in this energy range<sup>5</sup> and may be further corrected (i.e.,  $a = \phi(E_v)$ ) to give more exact variation of  $P(E_v^{+})$  over a range of energies; for the present  $a$  is retained as a constant. With these assumptions<sup>39</sup>

$$k_E = Q \frac{(n-1) \int_0^{E^{+}} [E_v^{+} + aE_z^{+}]^{n-2} [E^{+} - E_v^{+}] dE_v^{+}}{\int_0^E [E_v + E_z]^{n-1} dE_v}, \quad (\text{j})$$

$$Q = P_1^{+} P_R^{+} \prod_{i=1}^n h\nu_i / hP_1^{*} P_R^{*} \prod_{i=1}^{n-1} h\nu_i.$$

<sup>38</sup> The average hindering potential for the methyl groups is taken to be 1 kcal as compared with methyls in *cis*-butene-2, which has been estimated at  $\sim 500$  cal.<sup>37</sup> It is possible that 1 kcal is too low [K. Ito, J. Am. Chem. Soc. **75**, 2430 (1953)].

$\ddagger$  Decompositions such as these involving double bond formation represent a class of reactions in which the entropy of the complex could be lowered because of the disappearance of internal rotational degrees of freedom.

$\S$  In the addition of  $\text{CH}_3$  to *cis*- and *trans*-butene-2, it may be noted that inhibition of rotation about the central bond is found by Bader, Buckley, Leavitt, and Szwarc, J. Am. Chem. Soc. **79**, 5621 (1957).

<sup>39</sup> Upon integration (noting that  $E_z = 0$  at  $E = 0$ ) Eq. (j) becomes

$$k_E = Q[(E^{+} + aE_z^{+}) / (E + E_z)]^n.$$

Insertion of (i) and (j) into (b) and (c) with the substitutions  $\alpha = E_{\text{min}}^{+} + aE_z^{+}$ ,  $\beta = E_{\text{min}}^{+} + E_z$  gives

$$k_{a\infty} = \frac{Q \int_0^{\infty} [(E^{+'} + \alpha) / (E^{+'} + \beta)]^n \sum dE^{+'}}{\int_0^{\infty} \sum dE^{+'}}, \quad (\text{k})$$

and

$$k_{a0} = \frac{Q \int_0^{\infty} \sum dE^{+'}}{\int_0^{\infty} [(E^{+'} + \beta) / (E^{+'} + \alpha)]^n \sum dE^{+'}}, \quad (\text{l})$$

where

$$\sum = \sum_{E_v^{+'} \leq E^{+'}} P(E_v^{+'}) [E^{+'} - E_v^{+'}] \exp(-E^{+'}/RT) dE^{+'}$$

For the evaluation of (k) and (l) the ratio of  $P_1^{+}/P_1$  in the over-all rotations has been taken as unity; this lower limit cannot be seriously in error.<sup>1</sup> Also,  $P_R^{+}$  is set equal to  $P_R^{*}$  since the hindering potential for the methyl groups in the energized molecule is taken as  $\sim 1$  kcal, while in the complex the rotation barrier for the separating methyl drops, but that for the other methyl rises toward the value of 1.9 kcal for propylene.<sup>40</sup> If the value 1 kcal is too low, then the calculated value is also low. Of course, the  $Q$  factor cancels in any case in the ratio,  $k_{a\infty}/k_{a0}$ .

Since the factor  $a$ , set constant in Eq. (j) does not, quite give the correct energy dependence of  $P(E_v)$  in the range of energies  $E^{+} = 10$ –15 kcal, an empirical correction<sup>5</sup> evaluated from simple molecular models was inserted.

The comparison of observed and calculated values (Model I) is given in Table IV. Although the experimental values are given to two places of decimals, the scatter of the data in Figs. 3 and 4 indicates that the listed values merely give the order of magnitude;

TABLE IV. Comparison of observed and calculated values of  $k_a$  ratios.

|                                     | Temp. °C | Observed          | Model I | Model II |
|-------------------------------------|----------|-------------------|---------|----------|
| $k_{a\infty}(T_1)/k_{a\infty}(T_2)$ | 25/–78   | 1.70              | 1.73    | 1.06     |
| $k_{a0}(T_1)/k_{a0}(T_2)$           | –78/–103 | 1.26              | 1.36    | 1.06     |
|                                     |          | 1.30 <sup>a</sup> | 1.14    | 1.04     |
|                                     |          | 1.04 <sup>a</sup> | 1.05    | 1.03     |
| $k_{a\infty}/k_{a0}$                | 25       | 1.71              | 1.48    | 1.01     |
|                                     | –78      | 1.27              | 1.17    | 1.00     |
|                                     | –103     | 1.0 <sup>a</sup>  | 1.08    | 1.00     |
| $k_{a\infty}/Q(\alpha/\beta)^n$     | 25       |                   | 2.7     | 1.19     |
|                                     | –78      |                   | 1.5     | 1.12     |
|                                     | –103     |                   | 1.4     | 1.08     |
| $k_{a0}/Q(\alpha/\beta)^n$          | 25       |                   | 1.8     | 1.18     |
|                                     | –78      |                   | 1.3     | 1.11     |
|                                     | –103     |                   | 1.3     | 1.08     |

<sup>a</sup> Based on only two runs at –103°.

<sup>40</sup> D. R. Lide and D. E. Mann, J. Chem. Phys. **27**, 868 (1957).

this is adequate. The agreement both with respect to pressure and temperature variation is quite satisfactory. For illustrative purposes, the results of calculations are also given in which  $P(E_v^+)$  (but not the internal rotations) in the association complex was assumed to be unity (Model II). In contrast with conventional thermal experiments, the correspondence between this ostensibly unrealistic model and the observed results with respect to order of magnitude (and with respect to detail, also, at  $-103^\circ$ ) is quite striking. The small magnitude of  $k_{\text{aco}}/k_{\text{a0}}$ , just greater than unity, is many-fold smaller than is  $F$  (and  $G$ ) for thermal reactions of molecules of comparable complexity. This indicates the importance of quantum restrictions on excitation of vibrational degrees of freedom at these low temperatures and the consequent narrow range of energies (relative to  $E_{\text{min}}^+$  in particular) of the energized radicals.

$k_{\text{aco}}/k_{\text{a0}}$  approaches unity at the lowest temperature,  $-103^\circ$ . Since  $E_{\text{min}}^+$  is 10 kcal, the average value of  $P(E_v^+)$  is actually a large number even at  $-103^\circ$ . The concordance with Model II arises from the monoenergetic character of the energized species produced in this system. In that sense, this is an instance where the (semi-) classical Hinshelwood model holds. In Figs. 2-4, the "straight" line at  $-103^\circ$  reflects this situation, and the increasing curvature with rise of temperature indicates pictorially the loosening of quantum restrictions and increasing, although comparatively limited, departure from a constant value of the rate constant characterizing the energized molecules.

This is illustrated in another way in Table IV by the ratio  $k_a/Q[\alpha/\beta]^n$  where the denominator is the value of  $k_a$  if the radicals were precisely monoenergetic at the lowest vibrational energy. For  $k_{\text{a0}}$  at  $-103^\circ$  the ratio is only 1.3, illustrating that  $k_{\text{a0}}$ , which is equal to  $1/(1/k_E)$ , tends to  $1/k_{E_{\text{min}}}$  in general.<sup>7</sup>

The data provide a test also of the absolute magnitude of  $k_a$ . The value calculated for Model I at  $-103^\circ$  is  $k_a = 3.0 \times 10^6 \text{ sec}^{-1}$ ; the observed value is  $5.3 \times 10^6 \text{ sec}^{-1}$ . This temperature is convenient for comparison since there is negligible pressure effect and the rate constants correspond to those for radicals "monoenergetic" at  $E_{\text{min}}^+ = 10 \text{ kcal}$ . From Table IV, the absolute constant can be obtained for any conditions. It is evident that Model I affords satisfactory agreement with the observed values calculated with the collision diameters listed in Appendix III and on the assumption that deactivation occurs on every collision, thus making the values an upper limit.

Examination of different postulates reveals that although the absolute value of  $k_a$  is sensitive to assumptions about the activated complex for C—C rupture, it is fairly insensitive to the model of the activated complex for addition of H atoms to butene, while temperature and pressure effects similarly afford a weak test for assumptions about both complexes. This

circumstance arises in this case where the decomposition reaction is not the reverse of the activation association step and where the minimum energy for decomposition possessed by the energized radicals is appreciably greater than the thermal energy contributions involved. Thus variation by several hundred wave numbers of the assumed torsion frequency, or two C—H bending frequencies, in the *association* complex affects the calculated pressure and temperature ratios by only a few percent. In fact, in the limit of the "loose" association complex, replacement of the C—H bending modes by two degrees of rotational freedom has similar negligible effect. On the other hand, designation of certain degrees of freedom for the *decomposition* process as adiabatic raises the calculated absolute value of  $k_a$  appreciably; for three C—H stretching vibrations adiabatic,  $k_a = 4 \times 10^6 \text{ sec}^{-1}$ ; for three  $\text{CH}_3$  deformation vibrations adiabatic,  $k_a = 8 \times 10^6 \text{ sec}^{-1}$ . A larger number of adiabatic degrees of freedom (other than C—H stretch) would raise  $k_a$  excessively above the observed value. Similarly, if instead of taking all vibrations of the decomposition complex equal to those in butyl radical, the four vibrations (skeletal bend,  $\text{CH}_3$  rock (2),  $\text{CH}_2$  bend) that are lost on rupture (exclusive of internal rotation of methyl and the C—C stretch) are arbitrarily set for the complex at one-half their normal value, the calculated value of  $k_a$  rises by a factor of 6.5|| to  $1.9 \times 10^7 \text{ sec}^{-1}$ ; the effect on pressure and temperature ratios as in Table IV would amount to only a few percent, however. Use of a loose complex for C—C rupture would raise  $k_a$  by orders of magnitude.

Thus although Model I prescribes  $n = 31$  plus two internal rotations and a value as low as  $n \sim 20$ –25 may be equally valid, it appears that substantially all the vibrational degrees of freedom of the butyl radical come in play, as opposed to some lesser number such as  $n/2$ . Since the radicals are monoenergetic (with random energy distribution) direct comparison may be made with Slater's expression<sup>8</sup> for  $k_E$ , the classical form of which is  $k_E = \bar{\nu}(E - E_0/E)^n$ ; the point being that  $n$  is here the total number of active vibrational degrees of freedom (counting degenerate modes fully)<sup>3a</sup> and not the  $s$  of Kassel (which we will call "the falloff parameter") which is constrained to be<sup>41</sup>  $\leq (n + 1/2)$  in Slater theory. Thus our results appear to support the Slater definition of  $n$ , inasmuch as a falloff parameter as high 20–30 for this molecule would appear out of line with known behavior of unimolecular decomposition of molecules of related complexity. However, another aspect worth remarking on is that the present results tend to show that the different types of motions in the molecule (all) interact with the bond breaking coordinate. A similar situation prevails for the case of

|| Although the  $Q$  ratio in  $k_a$  is increased by 2<sup>4</sup>, a partially compensating decrease of  $E_s^+$  of 2.4 kcal occurs.

<sup>41</sup> N. B. Slater, Phil. Trans. Roy. Soc. (London) **246**, 57 (1953). The "less than" stipulation applies if the molecule possesses degenerate modes, which is not the case here.

ethyl radical<sup>10</sup> which, unlike the butyl case, involves formation and rupture of a C—H bond. Marcus found support for such a conclusion in the cases he analyzed. These include cases of C—H bond formation and rupture as well as C—H formation followed by C—C rupture. It may be questioned whether, considering experimental uncertainty, the existing data in the literature on falloff of unimolecular reactions<sup>42</sup> reveal other than the fact that substantially all vibrational modes are active in each case with the falloff parameter, however, following the Slater condition. We will not pursue the matter (or possible dilemma) any further here.

### Collisional Deactivation and Efficiency of Inert Gases

On a collision-per-collision basis the relative efficiencies,  $\eta$ , for deactivation compared to *cis*-butene-2, found for hydrogen and helium are 0.28 ( $H_2$ ) and 0.24 (He) at 25° and 0.40 ( $H_2$ ) at -80°. These values are averages over a range of  $S/D$ , but in view of the monoenergeticity of the radicals no variation in  $\eta$  would be expected.<sup>43</sup> The equality for hydrogen and helium apparently accords with the correlation offered by Volpe and Johnston<sup>43</sup> for nonpolar molecules, as well as the observations for these gases by Boudart and Dubois.<sup>20a</sup> And although chemical reaction between butyl and hydrogen is possible, one would conclude that the intermolecular potential governing this collision is of a general type and not of a specific chemical nature such as might apply for species which can react.<sup>44</sup> On the other hand the high relative efficiency of hydrogen and helium (in fact embarrassingly high for the argument made just below) as well as the pronounced *increase* in efficiency for hydrogen at lower temperatures suggest a specific chemical interaction. This is part of the more general problem of the greater efficiency of deactivating collisions at high levels of vibrational excitation, relative to efficiency at levels closer to the ground state. As pointed out by Johnston,<sup>43</sup> the collision efficiency for deactivation in unimolecular reactions cannot be uniquely unity for all gases, or even the parent gas, and has not been established as unity for any gas. However, the measurements by Boudart and Dubois<sup>20a</sup> of the amount of energy (3–5 kcal) transferred per collision in their system (at higher temperature than ours) by complex molecules, such as pentane, prompt the assumption that in the present study only a few collisions by butene are

necessary to remove the 10+ kcal required for deactivation, or at least to remove sufficient energy so that the molecules are effectively deactivated. In any case, the quoted observed  $k_a$  values are upper limits. The variations of the model for calculation discussed earlier, which almost invariably *raise* the calculated values, are thus in the wrong direction.

In the calculation of  $k_a$  ratios between two temperatures, the possible variation of the butene collision efficiency has been ignored. Also, if butyl radicals were deactivated stepwise with *small* decrements of energy involved, modification of the above treatment would be necessary; this is being considered. However, the small value of the ratio  $k_{acc}/k_{a0}$  argues against a large *effective* range of energies of the decomposing radicals, and hence against appreciable contributions to the rate arising from collisional transitional energy states,  $E^+$ , between 0 and  $E_{min}^+$ . An effort will be made in future to widen the working pressure range.

We note the usefulness of the present experimental arrangement for studying relative collisional deactivation efficiencies of inert gases with energized species which are closely monoenergetic, and which are accessible for study at much lower temperatures than in thermal investigations, as well as over a wide range of temperatures.

This work is being extended to other butenes as a source of the same *sec*-butyl radical in different states of excitation, to the use of D atoms and to other inert gases.

## APPENDIX I

### A. Radial Distribution Functions for Concentrations of Stable Species

#### (a) *cis*-Butene-2

Treating the rectangular plane slit as a hypothetical, spherical orifice (radius  $r_0$ ) of equivalent area, this problem is analogous to the calculation of the temperature of a hollow sphere ( $r_0 < r < r_w$ ) with radiation at the surfaces proportional to the first power of temperature.<sup>45</sup> The consumption of butene by reaction is small compared to loss by back-effusion at the orifice and is neglected.

The solution is

$$p_B(r) = (F_1/4\pi D_B) [(4D_B/c_B r_0^2) + (1/r_0) - (1/r)]$$

where  $D_B$  is the self-diffusion coefficient of butene,  $c_B$  the average speed of butene molecules, and  $F_1$  is the effusive flow out at  $r=r_0$ .  $p_B$  is a constant throughout the reactor except in the immediate vicinity of the orifice. The foregoing gives an overestimate of the pressure drop at the orifice, which is actually a slit of length  $25r_0$ , and the pressure drop in the worst (high-pressure) case would be only a few percent.

<sup>42</sup> As for cyclopropane,<sup>14</sup> nitrogen pentoxide [H. S. Johnston and R. L. Perrine, J. Am. Chem. Soc. **73**, 4782 (1951)], and nitril chloride, [H. F. Cordes and H. S. Johnston, J. Am. Chem. Soc. **76**, 4264 (1954)].

<sup>43</sup> M. Volpe and H. S. Johnston, J. Am. Chem. Soc. **78**, 3903 (1956).

<sup>44</sup> J. Franck and A. Eucken, Z. physik. Chem. **20B**, 460 (1933); K. F. Herzfeld, and V. Giffing, J. Phys. Chem. **61**, 844 (1957); K. Herzfeld, *Thermodynamic and Physics of Matter*, F. D. Rossini, editor (Princeton University Press, Princeton, New Jersey, 1955), Sec. H.

<sup>45</sup> Cf. H. S. Carslaw and J. C. Jaeger, *Conduction of Heat in Solids* (Clarendon Press, Oxford, 1947).

## (b) Propylene

This is the problem of steady-state temperature in a hollow sphere with radiation at the surfaces and heat generation throughout a function of  $r$ . Propylene formation rate is  $\rho \cdot$  (butyl radical formation) where  $\rho$  is determined from experiment and is a constant  $<1$ . Butyl radical formation  $= k p_H(r) p_B$ , where<sup>9</sup>

$$p_H(r) = A e^{-Cr}/r, \quad A = [b_0 e^{Cr}/4\pi D_H(1+Cr_0)],$$

$$C^2 = k p_B / D_H$$

$b_0$  is the rate of input of H atoms in pressure units,  $D_H$  is the diffusion coefficient of H atoms in butene, and  $p_B$  is taken constant. The solution is

$$p_P = -\frac{H}{C^2 r} e^{-Cr} + \frac{F_2}{4\pi D_P r} + \frac{H r_w e^{-Cr_w}}{Cr} \left[ 1 + \frac{1}{Cr_w} \right] + L,$$

where  $H = \rho k p_B A / D_P$ ,  $F_2$  is flow of propylene out at outer wall, and

$$L = (4HD_P e^{-Cr} / Cr_0 C_P) [1 + (1/Cr_0) + (C_P/4D_P C)]$$

$$- (F_2/4\pi D_P r_0) [(4D_P/C_P r_0) + 1]$$

$$- (H r_w e^{-Cr_w} / Cr_0) [(4D_P/C_P r_0) + 1] [1 + (1/Cr_w)].$$

$L$  is the dominant term and the maximum pressure drop across the reactor is  $<3\%$ .

## B. Radial Distribution Function for Concentrations of Butyl and Methyl Radicals

The steady-state equation of continuity for butyl radicals is

$$D_{Bu} \nabla^2 p_{Bu} - k_2 p_{Me} p_{Bu} - k_3 p_{Bu}^2 = -k p_H p_{Bu}.$$

Assume,  $p_{Me} = \delta p_{Bu}$ ,  $\delta =$  constant determined for each experiment. This assumption is reasonable in view of the similarity of the formation and removal processes for both methyl and butyl and is further justified below. Then for the case of spherical symmetry

$$Y''(r) + [2Y'(r)/r] - KY^2(r) = f(r),$$

where  $K =$  constant and  $f(r) = -Me^{-Cr}$ ,  $M =$  constant. Now the boundary conditions are not known since it is not known *a priori* whether butyl radicals get to the wall. Consider the case of the infinite reactor satisfying the conditions,

$$\lim_{r \rightarrow \infty} p_{Bu} = 0, \quad \text{and} \quad \int_{r_0}^{\infty} p_{Bu}^2 \cdot r^2 dr < \infty.$$

In this manner it can be determined if butyl gets to the wall in the actual reactor.

The reduced equation may be solved under these boundary conditions

$$W''(r) + [2W'(r)/r] - KW^2(r) = 0$$

by trial.  $W(r) = 2/Kr^2$  is a solution. Brownell<sup>46</sup> has given a good indication, through not a proof, that this is also the general solution.

Returning to the complete equation, insertion of the difference function  $U(r) = r[Y(r) - 2/Kr^2]$  yields the integral equation,

$$U(r) = - \int_r^{\infty} (r-s) [Me^{-Cs} - 4U(s)/s^2 - KU^2(s)/s] ds,$$

where  $U(r) = -Me^{-Cr}/C^2$  is an approximation to be iterated.

It is clear that the functional dependence of  $Y(r)$  of the complete equation cannot exceed that of the reduced equation and thus, without iteration, the desired information is obtained; butyl radicals get to the wall inasmuch as the volume of a spherical shell of the reactor depends upon  $r^2$ .

For the real reactor, reflection at the wall tends to reduce the radial dependence and  $Y(r) =$  constant is of plausible accuracy. The steady-state pressures of butyl and methyl were evaluated from the measured rates of production of octane, ethane, etc. For recombination of butyl, and of methyl+butyl, the specific rate constant was taken as the collision rate, with  $E=0$  and  $P$  factor  $=0.1$ , independent of pressure. At these pressures, methyl recombination may be in the falloff region. Dodd and Steacie<sup>47</sup> determined the falloff at 250°. The shift in falloff with temperature was calculated from Slater's equation<sup>41</sup> for  $\Delta \log p$  where  $n$  is set<sup>14</sup> equal to 11. Taking equal collisional efficiencies for acetone and *cis*-butene-2, the rate at  $-78^\circ$  is in the high pressure region above 0.2 mm, while at  $25^\circ$  the high-pressure limit is 0.8 mm. A constant value was used nonetheless at all pressures,  $k = 2 \times 10^{13}$  cc mole<sup>-1</sup> sec<sup>-1</sup>, since  $n$  is quite possibly  $>11$ .

Earlier  $p_{Me}$  was set equal to  $\delta p_{Bu}$ ,  $\delta$  constant. For  $p_R \sim$  constant, this is valid. A check can be made with the aid of the Schwarz inequality, whence

$$\frac{R(\text{ethane}) \cdot R(\text{octane})}{R^2(\text{isopentane})} \geq \frac{k_{C_2H_6} \cdot k_{C_8H_{18}}}{k_{i_{80}-C_5H_{12}}^2}.$$

On substitution of the average measured rate ratios and estimated rate constants, there is obtained  $0.37 >$

<sup>46</sup> F. H. Brownell (private communication). We thank Professor Brownell of the Mathematics department for valuable discussion of this equation.

<sup>47</sup> R. E. Dodd and E. W. R. Steacie, Proc. Roy. Soc. (London) **A223**, 285 (1954).

0.30. In view of uncertainties, the assumption  $\delta = \text{constant}$  is as good as any.

#### APPENDIX II

The following frequencies ( $\text{cm}^{-1}$ ) for butyl were mainly transferred from those for butane.<sup>48</sup> Since these frequencies enter through the semiclassical equation for  $P(E_v)$ , the exact details of the assignment as well as modest variation of the frequencies are of no consequence:

C—H stretch, 2950 (9); C—C stretch, 950 (3);  
CH<sub>2</sub> rock, 750; CH<sub>2</sub> twist, 1300; CH<sub>2</sub> wag, 1340;  
CH<sub>2</sub> bend, 1450; CH<sub>3</sub> unsym. deform., 1450 (4);  
CH<sub>3</sub> sym. deform., 1380 (2); CH<sub>3</sub> rock, 950, 1040 (3),

<sup>48</sup> Szasz, Sheppard, and Rank, *J. Chem. Phys.* **26**, 704 (1948);  
R. A. Rasmussen, *ibid.* **16**, 712 (1948); K. S. Pitzer, *ibid.* **12**,  
310 (1944).

CH wag, 1140 (2); skeletal bend, 320 (2); skeletal  
torsion, 250 (assumed).

#### APPENDIX III. COLLISION CROSS SECTIONS USED

| Species              | $10^8\sigma$ , cm | Source                                                                                         |
|----------------------|-------------------|------------------------------------------------------------------------------------------------|
| H, D atom            | 2.7               | Reference 13                                                                                   |
| Hydrogen             | 2.7               | E. H. Kennard, <i>Kinetic Theory of Gases</i> (McGraw-Hill Book Company, Inc., New York, 1938) |
| Methyl               | 3.5               | G. B. Kistiakowsky and E. K. Roberts, <i>J. Chem. Phys.</i> <b>21</b> , 1637 (1953)            |
| <i>cis</i> -Butene-2 | 6.5               | Reference 13                                                                                   |
| <i>sec</i> -Butyl    | 6.5               | Assumed                                                                                        |
| Helium               | 2.2               | Kennard                                                                                        |